INSTITUTE OF ENGINEERING & MANAGEMENT, KOLKATA

Department of Computer Science

LAB 3 HLD

System Modelling (UML/DFD) and UI Prototyping

Software Engineering and Project Management Laboratory | Course Code: PCCCS594 | Semester V

Team Name / Group No.: 404 Brain Not Found / 10

Project Title: IEM Admission Portal

Team Members (Name — Roll No.)

1.: Debarghaya Mitra – 248

2.: Arka Ghosh Dastider – 220

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5.: Jatin Kumar – 232

Section / Batch: C / 2024 – 2028 / 5th Semester / 3rd Year

Date: 21.06.2026

Faculty Signature:

How to Use This Worksheet

Fill in each section below **during the lab session**, working on your own project (not the library example from the Lab Manual). For every diagram: build it in draw.io or Figma as instructed in the **Lab 3 Manual**, export/screenshot it, and paste it into the matching dashed box. Tick each checklist item once you have verified it. Keep the Lab 4 Manual open beside you for the notation tables and step-by-step instructions.

0. YOUR PROBLEM STATEMENT

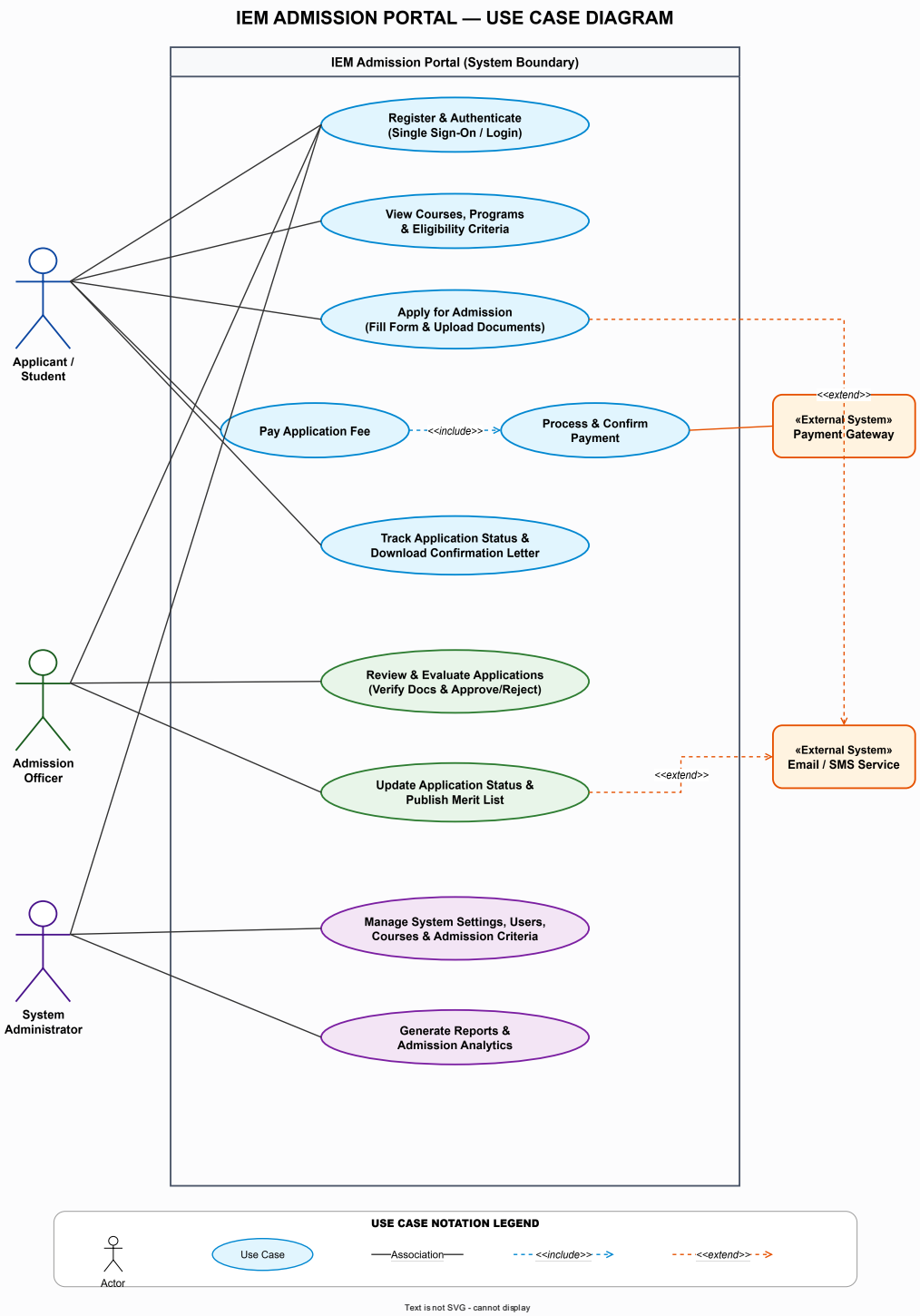
Write the one use case you are modelling in this worksheet, in 2–3 sentences

The **IEM Admission Portal** allows students to register, fill out and submit their admission applications, upload required documents, and pay the admission fee online. The system validates applications, enables admission officers to review them, and provides applicants with application status updates and confirmation notifications.

1. Use-Case Diagram

Required elements — tick once each is present in your diagram:

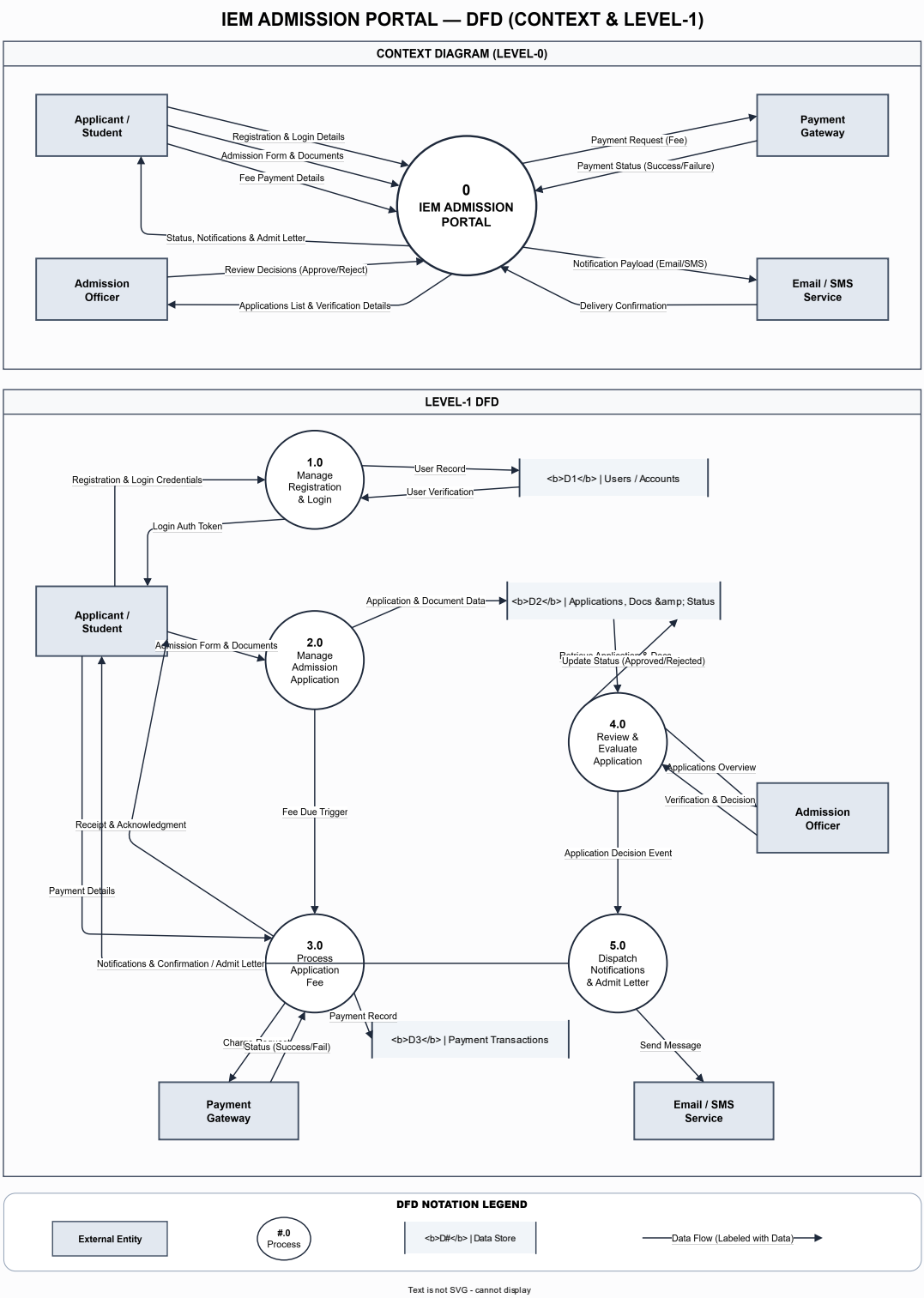
- ☑ All actors identified and labelled
- ☑ System boundary drawn and labelled
- ☑ Every use case is a distinct piece of functionality
- ☑ Associations connect the correct actor to the correct use case(s)
- ☑ Any «include» / «extend» relationship is labelled



2. DFD (Context + Level-1)

Required elements — tick once each is present in your diagram:

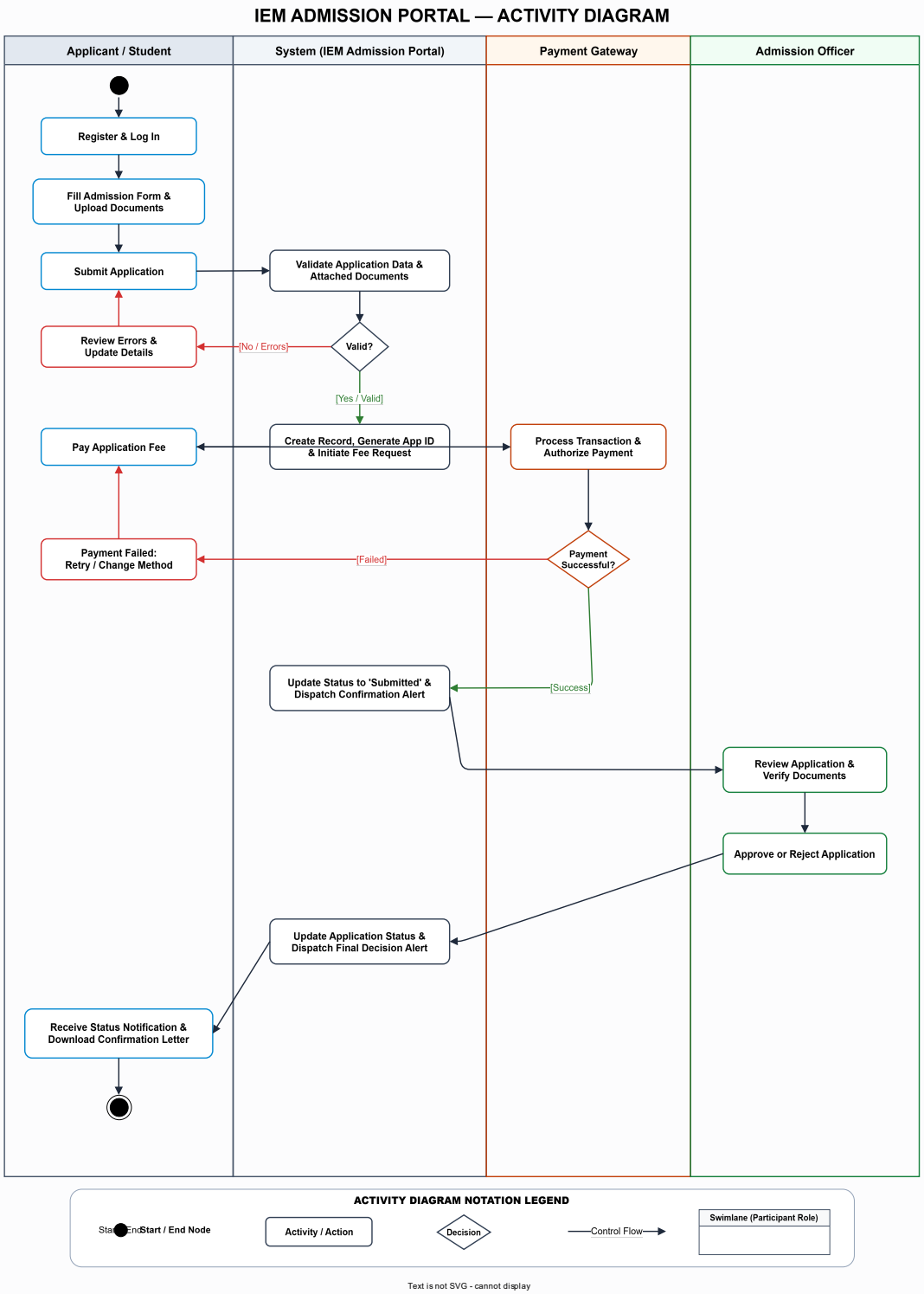
- ☑ Context diagram: whole system as ONE process, external entities shown, no data stores
- ☑ Level-1 diagram: numbered processes, data stores, correct symbol for each
- ☑ Every context-level flow reappears in the Level 1 diagram (levelling rule)
- ☑ All data flows are labelled



3. Activity Diagram

Required elements — tick once each is present in your diagram:

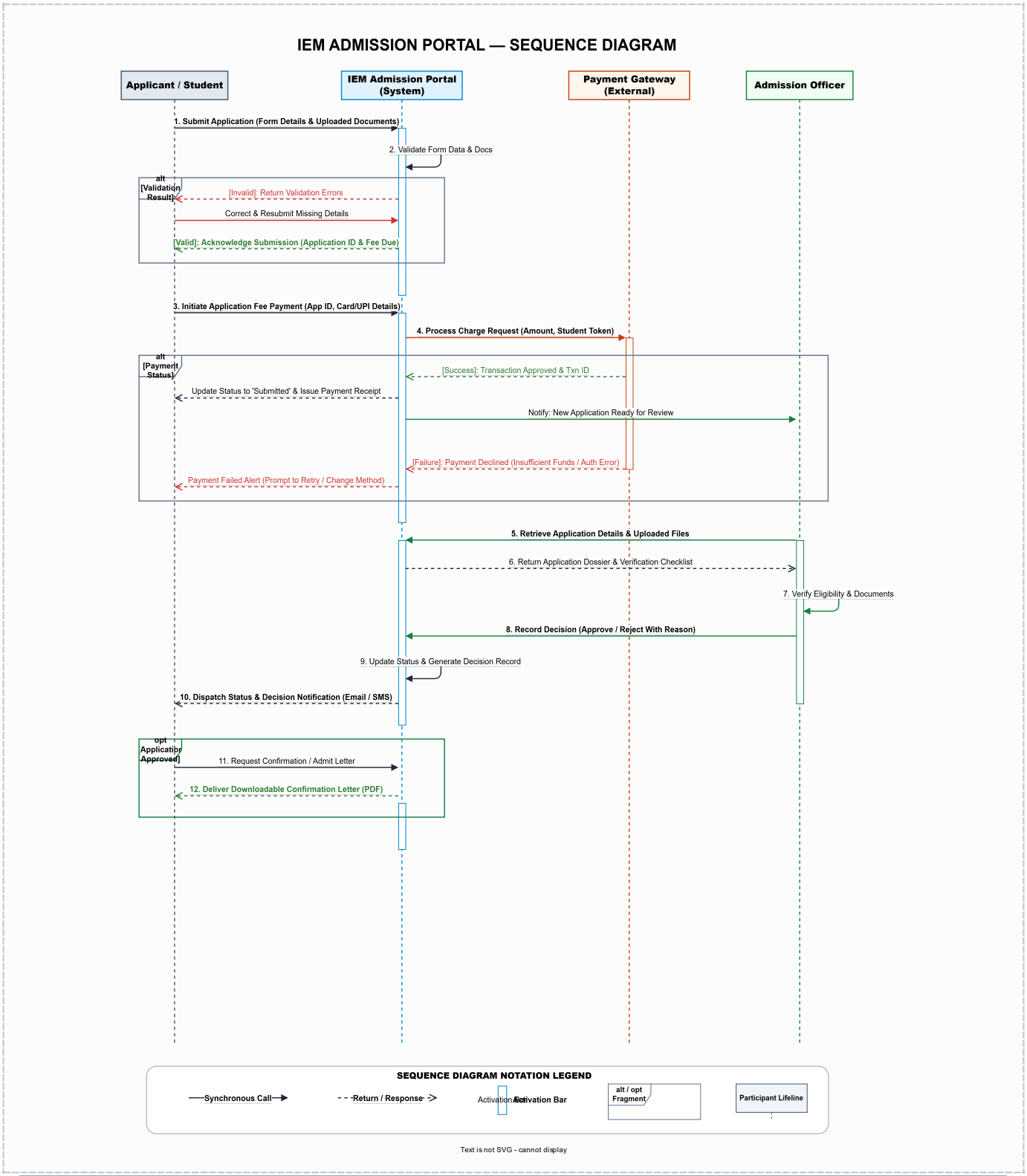
- ☑ Initial node and final node present
- ☑ Every decision point uses a diamond with labelled branches
- ☑ Actions are clearly named (verb + object)
- ☑ (Optional) Redrawn with swimlanes showing responsibility



4. Sequence Diagram

Required elements — tick once each is present in your diagram:

- ☑ One lifeline per participating actor/object, correctly named
- ☑ Messages numbered in call order, top to bottom
- ☑ Synchronous calls drawn as solid arrows; returns as dashed arrows
- ☑ Activation bars shown on lifelines that are actively processing



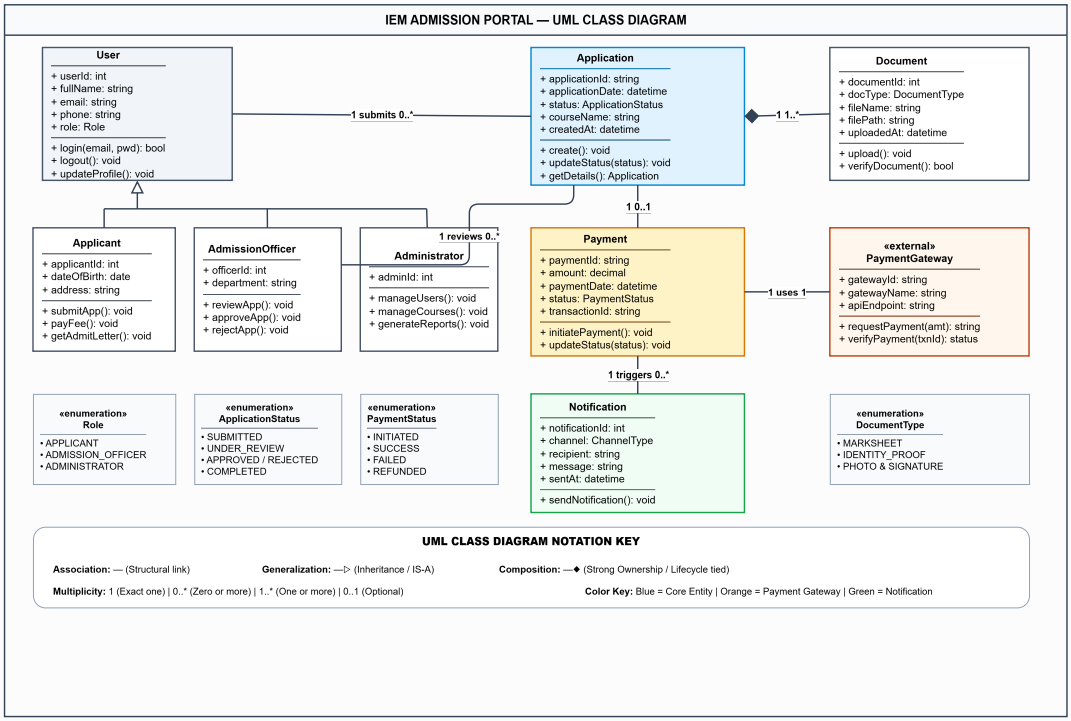
5. Design Diagram — UML Class Diagram / Structure Chart

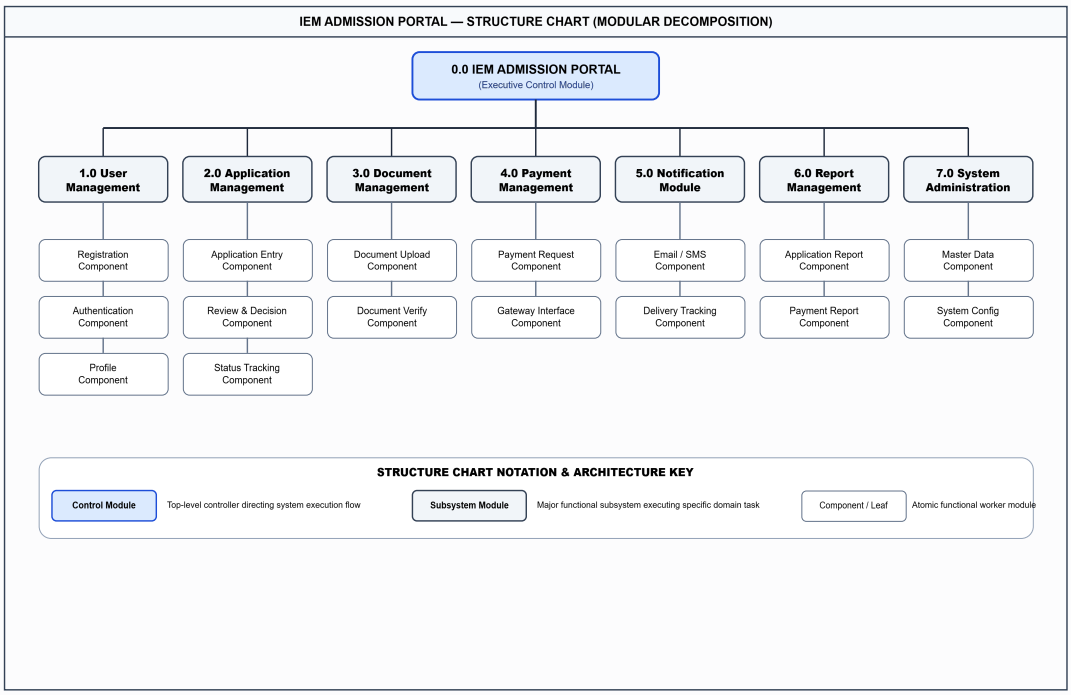
Required elements — tick once each is present in your diagram:

☑ Every class shows name / attributes / operations (or every module shown as a rectangle, for a structure chart)

☑ Relationships use the correct arrow/line style (association, inheritance, aggregation, composition / data & control couples)

☑ Multiplicities shown on associations (class diagram) or couples labelled (structure chart)



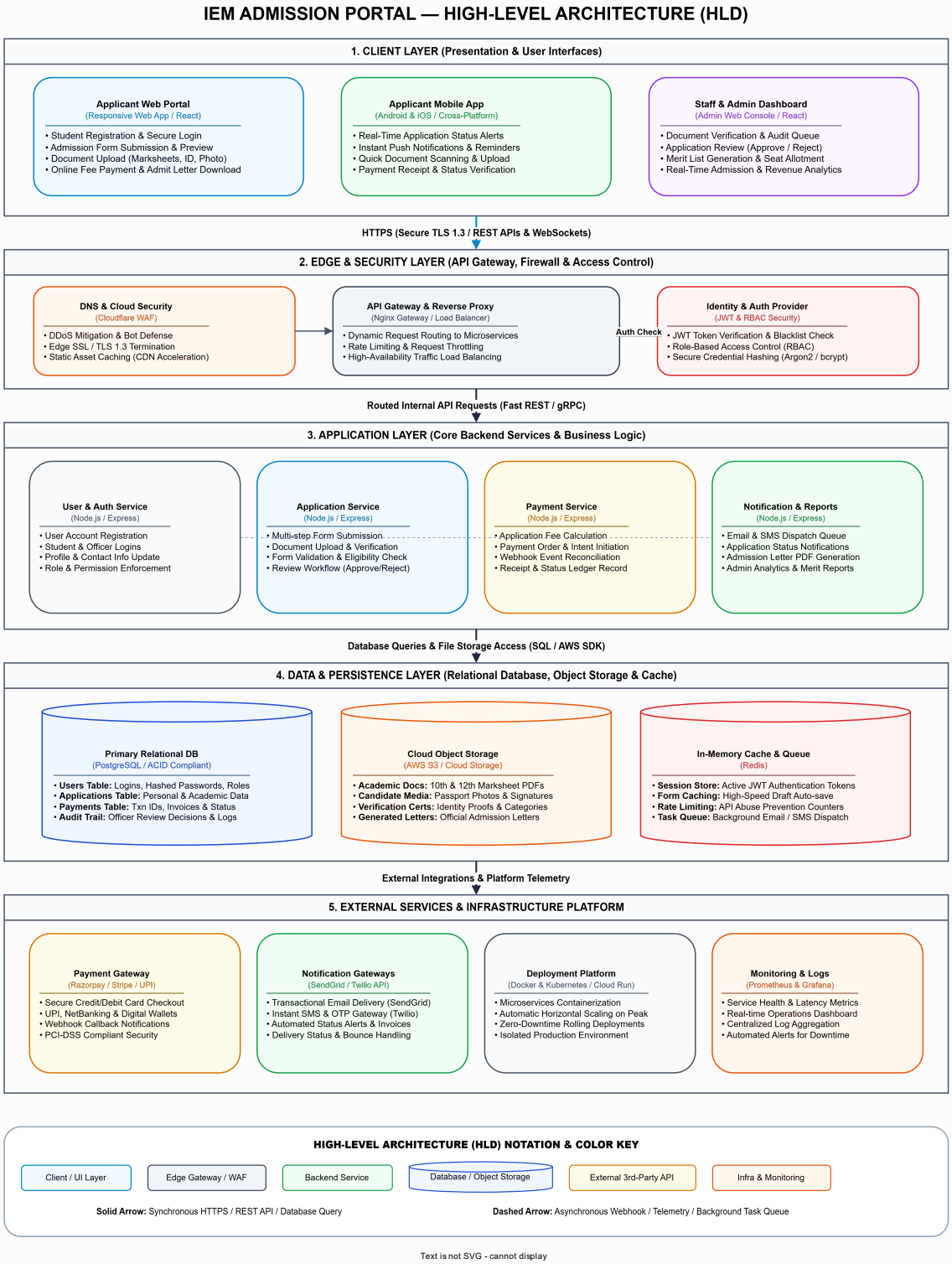


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5b. HLD / Architecture Diagram

Required elements — tick once each is present in your diagram:

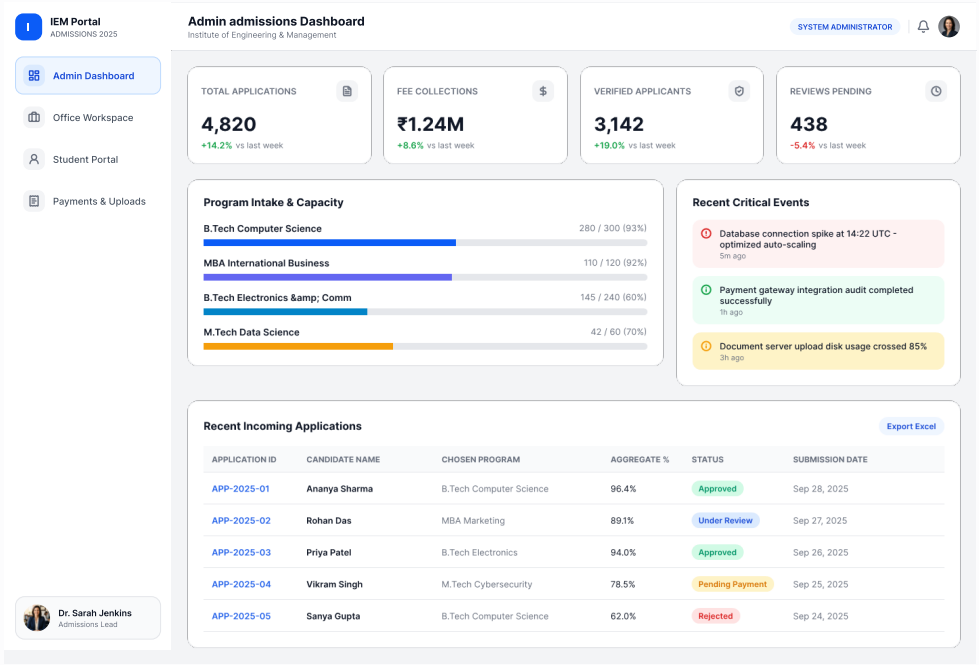
- ☑ Client, Application and Data layers (or your architecture's equivalent) clearly separated
- ☑ Every major module from your SRS appears as one component
- ☑ Databases / external systems shown with the correct symbol
- ☑ Connections between layers/components are labelled



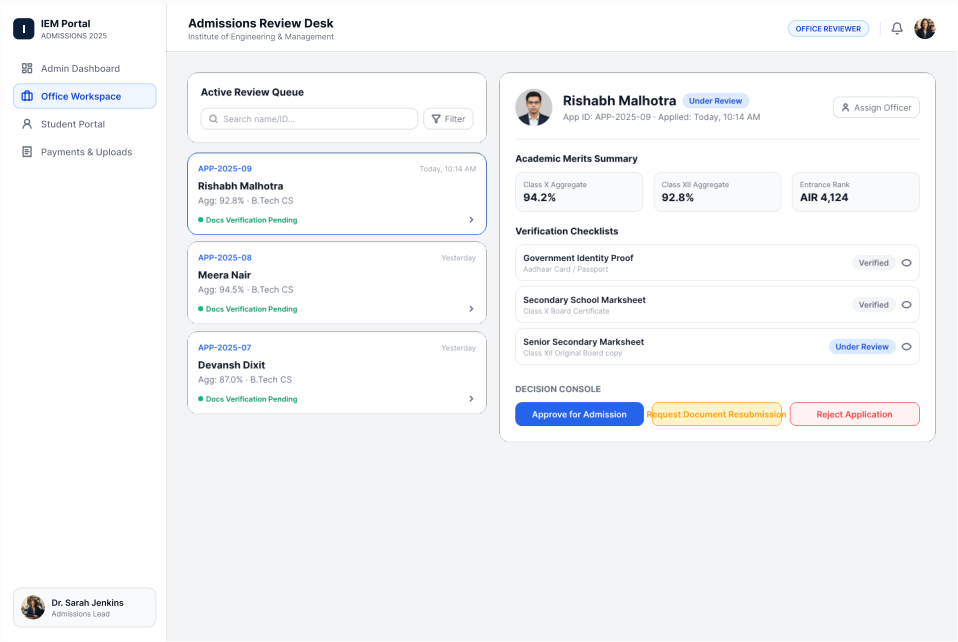
6. UI WIREFRAMES (Figma)

Paste a screenshot of each of your 3–4 core screens below (one box per screen). Label each with its screen name.

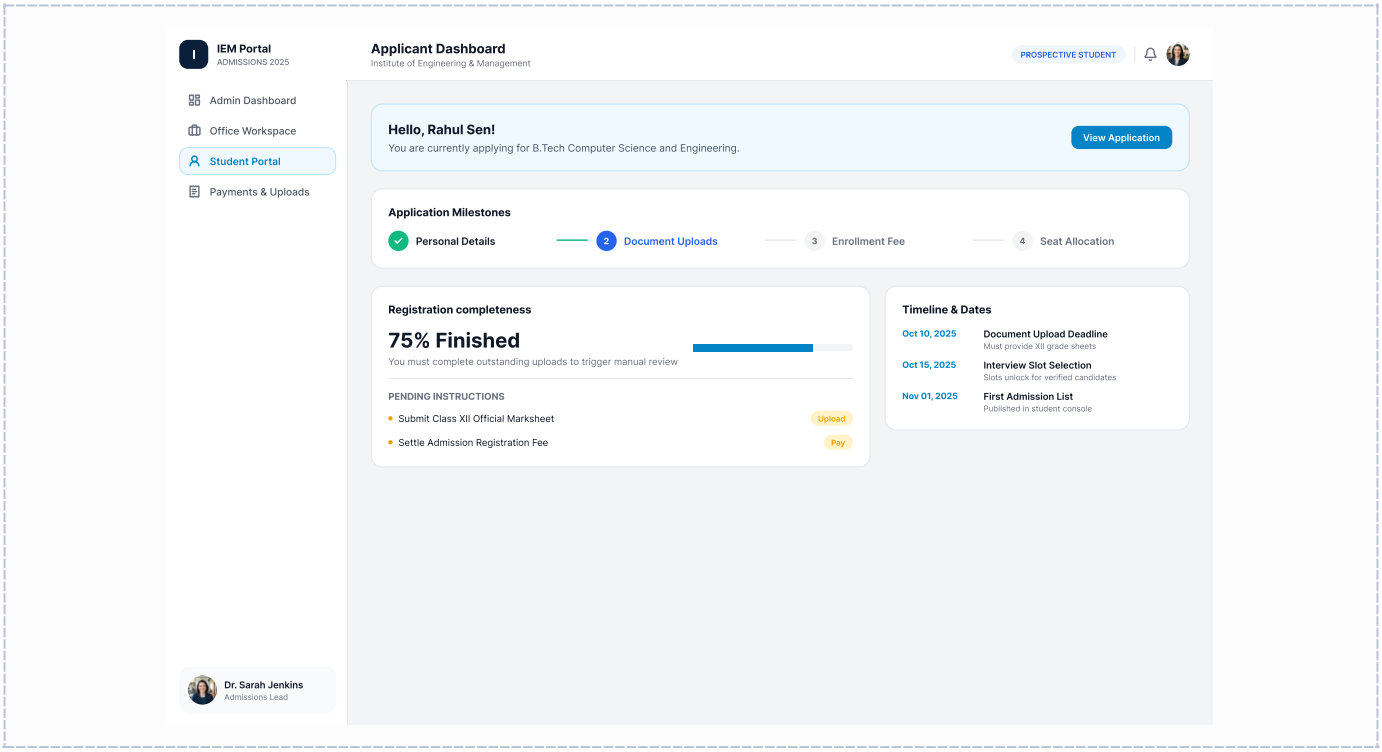
Screen 1 name: Admin Dashboard



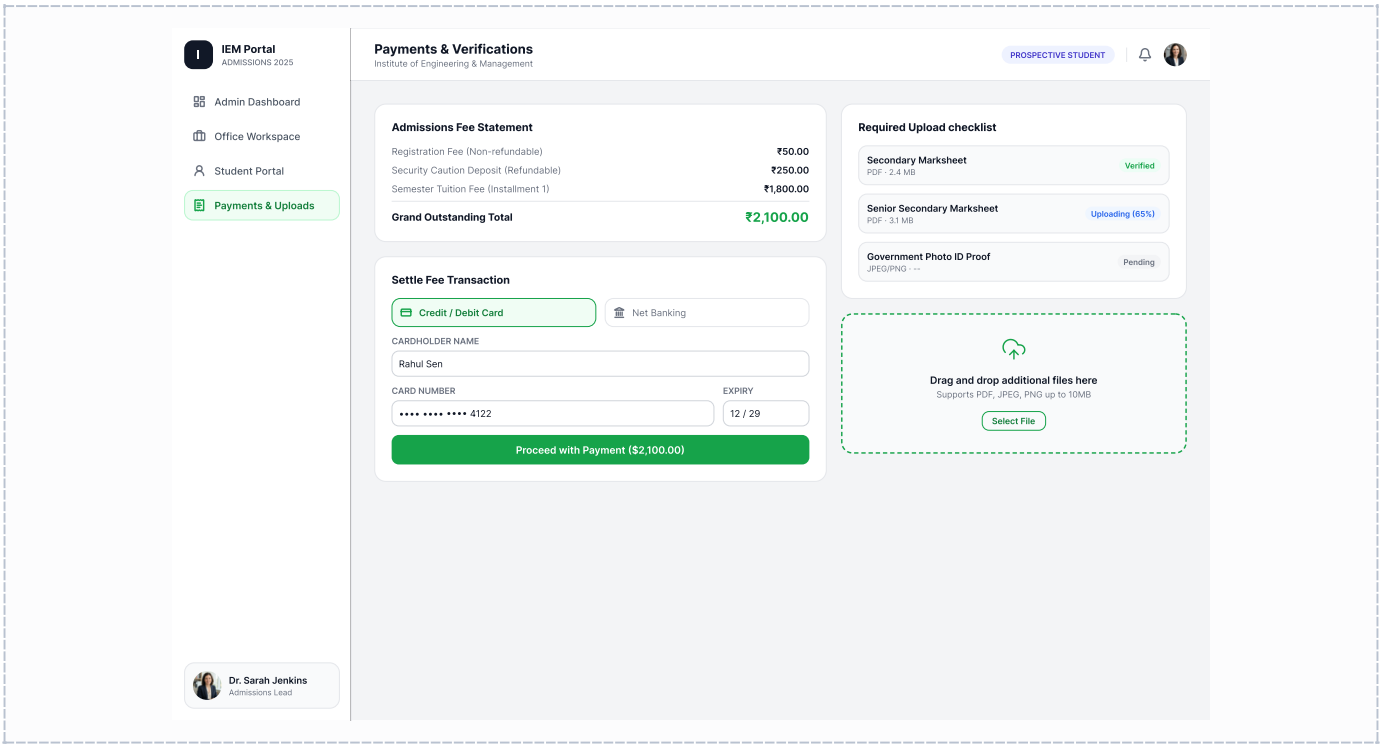
Screen 2 name: Admission Officer Portal



Screen 3 name: Student Portal



Screen 4 name: Document Upload and Payment Portal



Figma "Share prototype" link: https://www.figma.com/design/wCMMNYMxhcjZFkoRPKgYZy/IEM-Admission?node-id=0-1&t=C0YtsMXrr29OilGM-1

7. SELF-ASSESSMENT CHECKLIST

- ☑ Use-case diagram (actors + use cases, correctly notated)
- ☑ DFD — context diagram AND Level-1 diagram
- ☑ Activity diagram for the chosen use case
- ☑ Sequence diagram for the chosen use case
- ☑ UML class diagram (or structure chart) as the design diagram
- ☑ HLD / architecture diagram
- ☑ 3–4 Figma wireframe screens, grayscale, clearly labelled
- ☑ Figma frames linked into a working clickable flow
- ☑ Figma “Share prototype” link copied above (access set to “Anyone with the link — view”)
- ☑ Every diagram uses consistent actor/object names throughout
- ☑ Every team member's name and roll number on the cover page

For Faculty Use

Criterion	Marks Awarded	Remarks
Correct notation (25%)		
Completeness (20%)		
Consistency across diagrams (20%)		
Wireframe clarity & flow (20%)		
Presentation & labelling (15%)		
TOTAL		